

UNNS Color Confinement

Repair Threshold Report

Second Confinement Diagnostic - static-source gaps around reported string-breaking markers

Project: UNNS + Color Confinement Regime Synthesis

Status: formal diagnostic report; structural mapping, not a derivation of QCD confinement

Executive statement

The diagnostic formalizes the threshold window around the reported light and strange string-breaking markers. It treats the reported thresholds as repair-threshold markers: the stretched color route does not externalize free color; instead, it enters a channel-competition window where screened/color-neutral composite routes become the admissible continuation.

1. Purpose

This report formalizes the second diagnostic of the UNNS + color confinement investigation. The goal is to isolate the threshold region where a stretched color route stops being treated merely as increasing separation and becomes a channel-reorganization problem.

```
r near r_c  ~= 1.224 fm
r near r_cs ~= 1.293 fm
gap behavior
two-meson threshold interpretation
transition from stretched route to repaired composite route
```

2. Data boundary

The static-source levels used here are explicitly marked as `MODEL_RECONSTRUCTION_FROM_REPORTED_PARAMETERS_NOT_RAW_DATA`. They are useful for a controlled structural diagnostic, but they are not raw measured GEVP lattice points.

Quantity	Value
Static rows total	57
Static separation range	0.70686 to 1.60650 fm
Rows inside threshold-analysis window	17
Minimum global V1-V0 gap	0.07838 GeV at $r = 1.60650$ fm
Minimum global V2-V1 gap	0.04092 GeV at $r = 1.25307$ fm

Boundary statement: This diagnostic does not claim that UNNS derives QCD confinement. It converts the reported string-breaking region into a structured UNNS repair-window object.

3. Reported repair-threshold markers

Symbol	Channel	Reported r [fm]	Nearest reconstructed r [fm]	$V1-V0$ [GeV]	$V2-V1$ [GeV]
r_c	light_static_light_threshold	1.224	1.22094	0.08970	0.04594
r_{cs}	strange_static_strange_threshold	1.293	1.28520	0.08642	0.04639

The two reported thresholds are treated as repair-threshold markers, not as free-color externalization points.

4. Figures

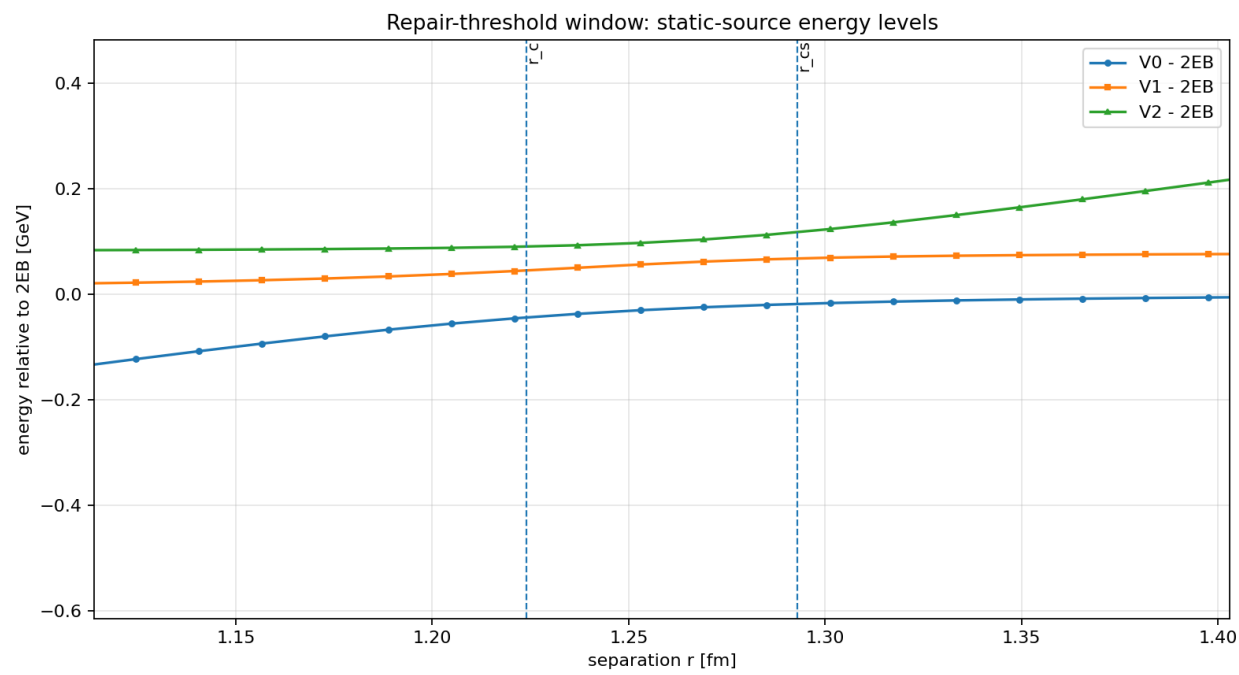


Figure 1. Repair-threshold energy-level window. Vertical markers indicate the reported light and strange threshold positions.

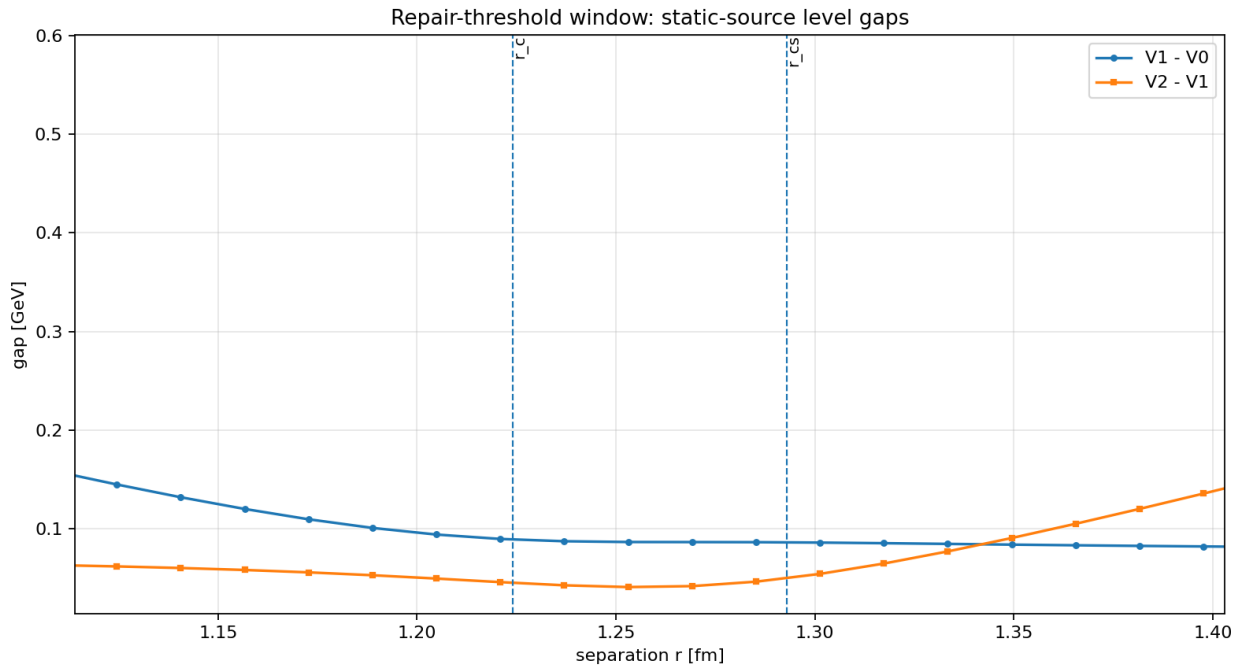


Figure 2. Repair-threshold static-source gap window. The gaps serve as channel-competition indicators in the repair-window region.

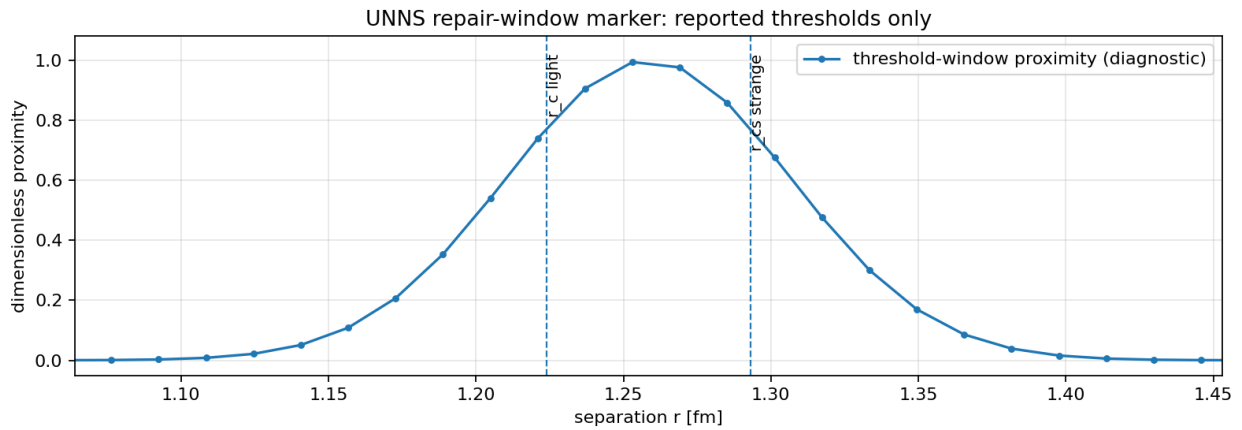


Figure 3. UNNS repair-window marker based on reported threshold proximity. This is a diagnostic marker, not a fitted law.

5. Diagnostic finding

The reconstructed spectrum places both reported string-breaking thresholds inside a region where the low-lying static-source gaps are compressed relative to the smaller-separation side. This makes the region suitable as the first UNNS repair-window object:

```

stretched color route
-> threshold-window approach
-> competing screened channel
-> repaired admissible color-neutral composite route

```

The relevant diagnostic is not simply that an energy rises. The relevant diagnostic is that the route-extension coordinate enters a channel-reorganization window where the forbidden output is not an isolated colored constituent but a screened or color-neutral composite channel.

6. UNNS object extracted

Physics item	UNNS interpretation
separation r	route-extension coordinate
V_0, V_1, V_2 static spectrum	boundary-pressure / route-tension spectrum
$V_1 - V_0$ and $V_2 - V_1$ gaps	channel-competition / repair-window indicators
r_c	light-channel repair-threshold marker
r_{cs}	strange-channel repair-threshold marker
screened two-meson channel	repaired admissible composite route

7. Interpretation

The second diagnostic transforms the broad color-confinement statement into a more precise UNNS regime object. The system is not interpreted as moving from bound color to free external color. Instead, the route-extension coordinate approaches a repair window in which competing screened channels become the admissible continuation.

```

attempted external separation
-> route tension
-> repair-threshold window
-> screened/color-neutral composite continuation

```

This is the first formal threshold-local expression of color confinement as admissibility-by-closure.

8. What is supported

- a concrete repair-threshold window around the reported light and strange string-breaking distances;
- a gap-based channel-competition diagnostic in that window;
- a clean UNNS mapping from route extension to threshold repair;
- preservation of the internal/external distinction: colored constituents remain internal coordinates, while external admissibility appears through color-neutral screened channels.

9. What is not yet supported

- a full measured-point analysis of $V_0(r)$, $V_1(r)$, $V_2(r)$;
- covariance/uncertainty propagation for the reconstructed spectrum;
- a fitted UNNS boundary-pressure law;
- a derivation of QCD confinement from UNNS.

10. Next required data/action

- Obtain author or digitized measured V_0, V_1, V_2 points around r_c and r_{cs} .
- Extend Baker flux-tube profiles across separations approaching the onset of string breaking.
- Build the first threshold-local boundary-pressure proxy only after measured or digitized static points are available.

Immediate next research question: Does route localization, measured through the chromoelectric flux-tube profile, change systematically as the static-source separation approaches the repair-threshold window?